



THE UNIVERSITY OF
MELBOURNE

Machine Learning Applications for Health

COMP90089 (2026) - Lecture 2
Prof Daniel Capurro





Today's Agenda

- **Quick Recap**
- **Open discussion**



What you reviewed before class

Introduction to Biomedical Informatics

Learning Healthcare System

What is machine learning?



Sergios Theodoridis

“Machine learning is a name that is gaining popularity as an umbrella and evolution for methods that have been studied and developed for many decades in different scientific communities and under different names, such as statistical learning, statistical signal processing, pattern recognition, adaptive signal processing, image processing and analysis, system identification and control, data mining and information retrieval, computer vision, and computational learning....

What is machine learning?



Sergios Theodoridis

“The name “machine learning” indicates what all these disciplines have in common, that is, to *learn from data*, and then *make predictions*. What one tries to learn from data is their underlying structure and regularities, via the development of a *model*, which can then be used to provide predictions.”

(Machine Learning: A Bayesian and Optimization Perspective, xiii)



ML and health informatics

Machine learning in health is a part of health informatics (medical informatics, biomedical informatics)

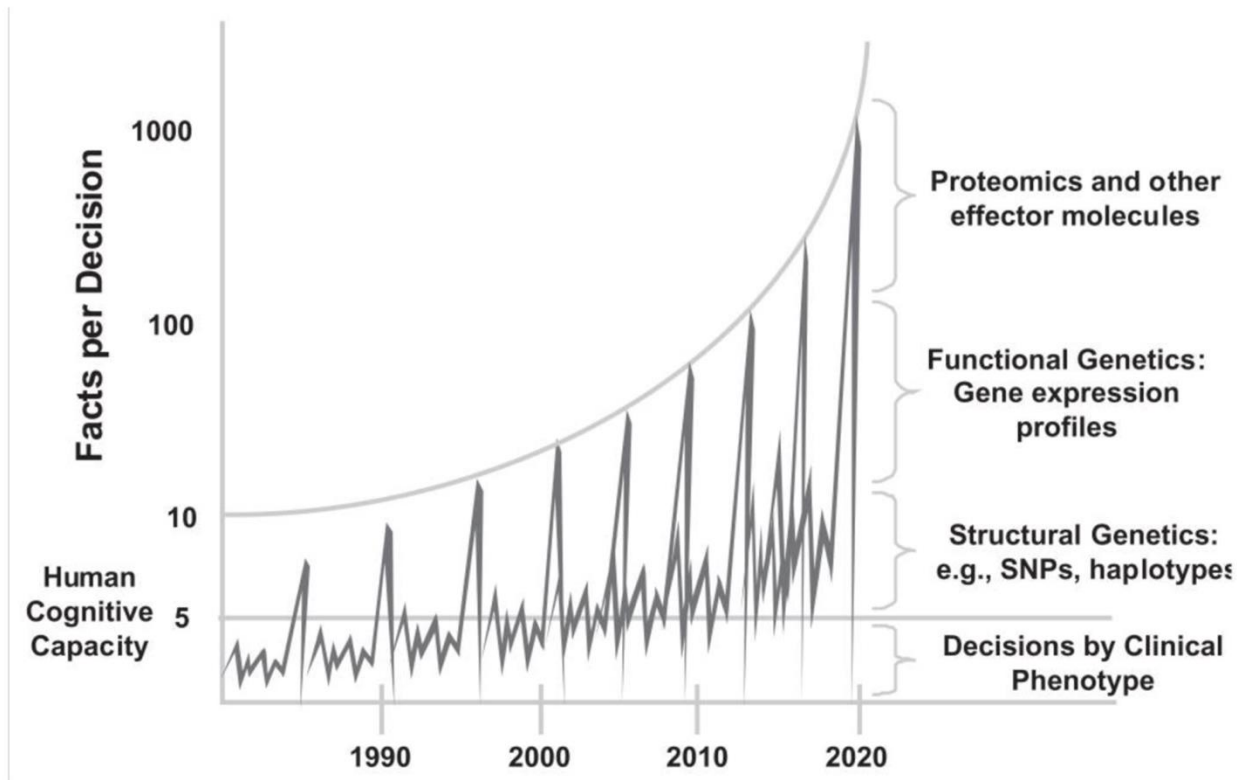


What is informatics?

“Informatics is the science of information. *It studies the representation, processing, and communication of information in natural and artificial systems.* Since computers, individuals and organizations all process information, informatics has computational, cognitive and social aspects.

“Used as a compound, in conjunction with the name of a discipline, as in medical informatics, bio-informatics, etc., it denotes the specialization of informatics to the management and processing of data, information and knowledge in the named discipline.”

Why is health informatics important?



Evidence-Based Medicine and the Changing Nature of Healthcare: 2007 IOM Annual Meeting Summary. Institute of Medicine (US).

Why is health informatics important?

The three numbers you need to know about healthcare: the 60-30-10 Challenge

Jeffrey Braithwaite^{1*}, Paul Glasziou² and Johanna Westbrook³

Braithwaite et al. *BMC Medicine* (2020) 18:102
<https://doi.org/10.1186/s12916-020-01563-4>

Received: 30 July 2019 Revised: 11 March 2020
Accepted: 17 March 2020 Published online: 04 May 2020

Abstract

Background: Healthcare represents a paradox. While change is everywhere, performance has flatlined: 60% of care on average is in line with evidence- or consensus-based guidelines, 30% is some form of waste or of low value, and 10% is harm. The 60-30-10 Challenge has persisted for three decades.

Main body: Current top-down or chain-logic strategies to address this problem, based essentially on linear models of change and relying on policies, hierarchies, and standardisation, have proven insufficient. Instead, we need to marry ideas drawn from complexity science and continuous improvement with proposals for creating a deep learning health system. This dynamic learning model has the potential to assemble relevant information including patients' histories, and clinical, patient, laboratory, and cost data for improved decision-making in real time, or close to real time. If we get it right, the learning health system will contribute to care being more evidence-based and less wasteful and harmful. It will need a purpose-designed digital backbone and infrastructure, apply artificial intelligence to support diagnosis and treatment options, harness genomic and other new data types, and create informed discussions of options between patients, families, and clinicians. While there will be many variants of the model, learning health systems will need to spread, and be encouraged to do so, principally through diffusion of innovation models and local adaptations.

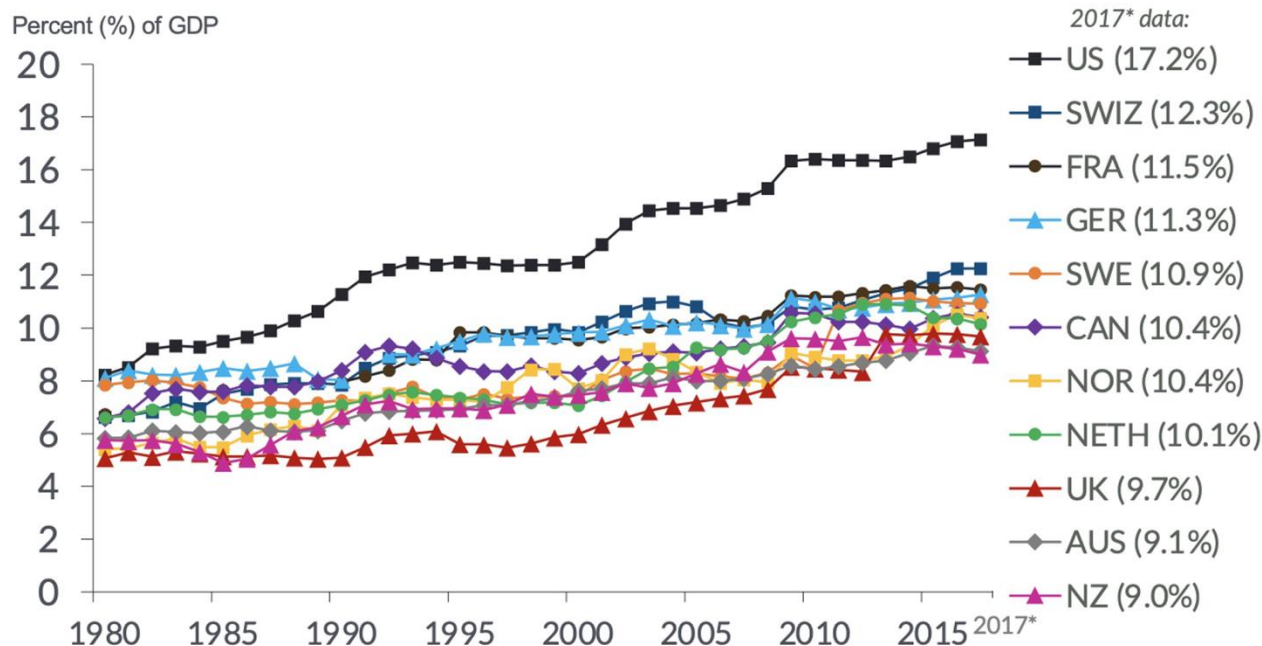
Conclusion: Deep learning systems can enable us to better exploit expanding health datasets including traditional and newer forms of big and smaller-scale data, e.g. genomics and cost information, and incorporate patient preferences into decision-making. As we envisage it, a deep learning system will support healthcare's desire to continually improve, and make gains on the 60-30-10 dimensions. All modern health systems are awash with data, but it is only recently that we have been able to bring this together, operationalised, and turned into useful information by which to make more intelligent, timely decisions than in the past.

Keywords: Learning health system, Complexity, Complexity science, Change, Evidence-based care, Clinical networks, Quality of care, Patient safety, Policy, Healthcare systems



Health Care Spending as a Percent of GDP, 1980–2017

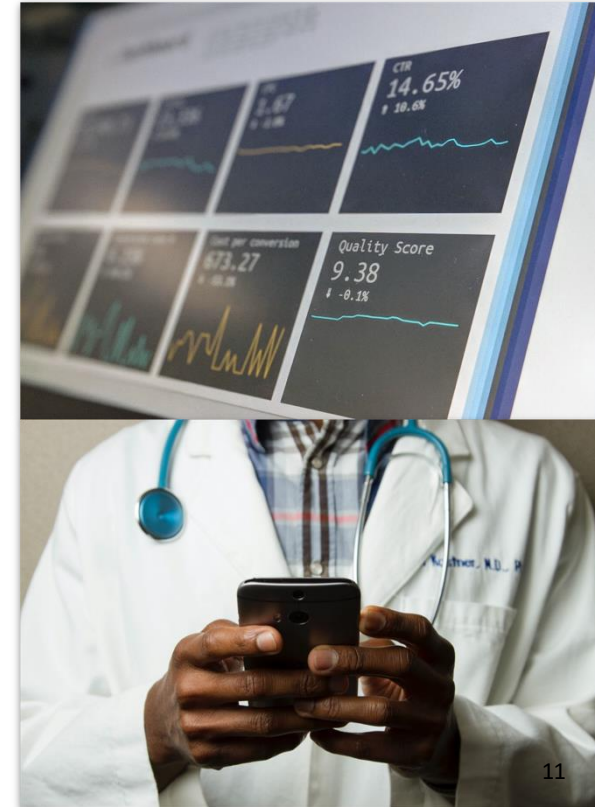
Adjusted for Differences in Cost of Living



Key Concepts

What informatics “is and isn’t” (Friedman, 2013)

- Is
 - Cross-training where basic informational sciences meet a biomedical application domain
 - Relentless pursuit of assisting people
 - Tower of achievement
 - Model formulation
 - System development
 - System implementation
 - Study of effects
- Isn’t
 - Scientists or clinicians tinkering with computers
 - Analysis of large data sets per se
 - Circumscribed roles related to deployment of electronic health records (*point of disagreement)
 - Profession of health information management
 - Anything done using a computer



Key Concepts

It has a “fundamental theorem” and a “golden rule”

Fundamental Theorem
(Friedman, 2009) – based on
“relentless pursuit of assisting
people”

Goal of informatics is

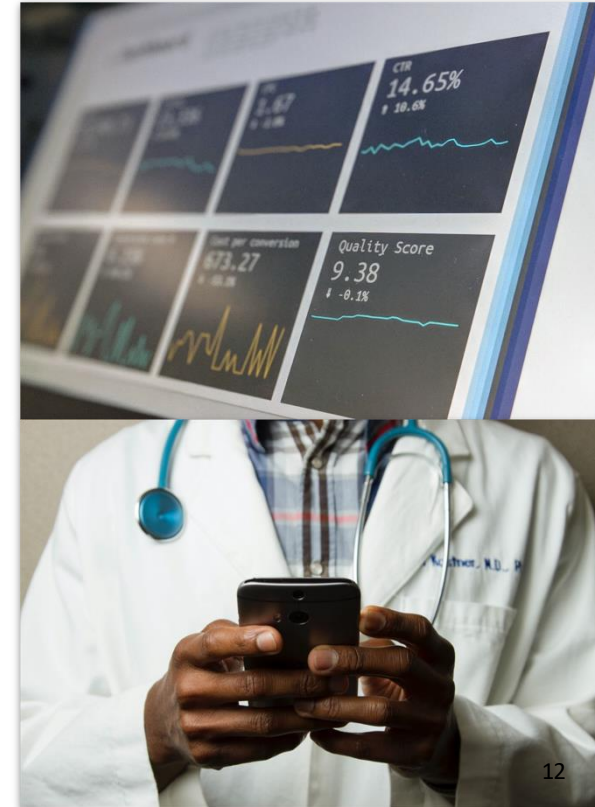
$$\left(\text{brain} + \text{computer} \right) > \text{brain}$$

Goal is not

$$\text{computer} > \text{brain}$$

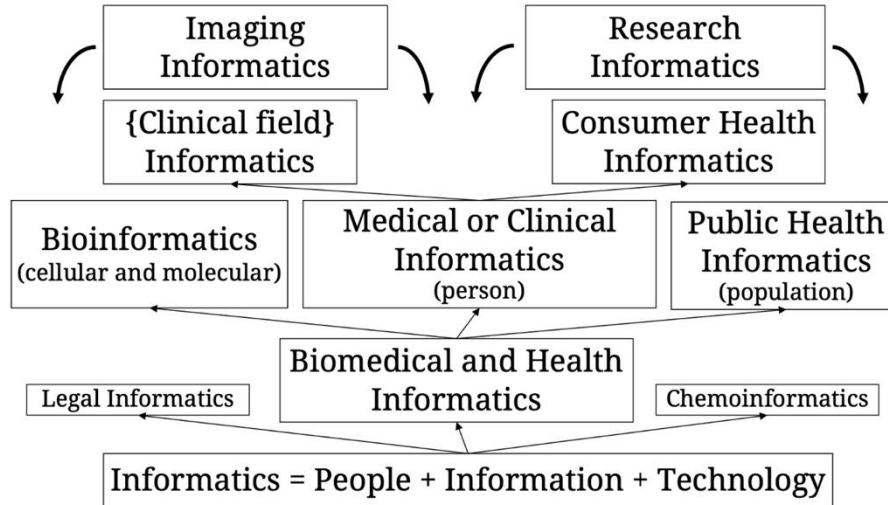
Golden Rule
(Kuperman, personal
communication,
2013):

“Never implement
unto others that
which you would not
implement unto
yourself”

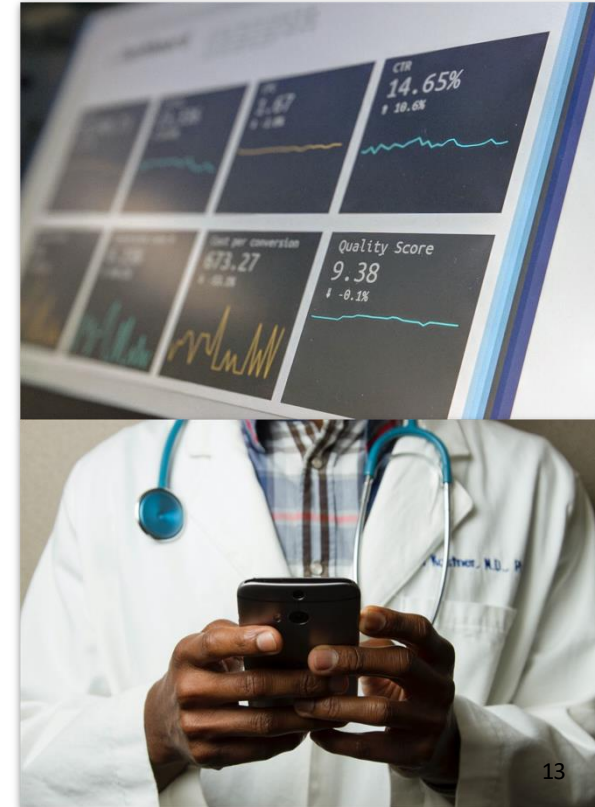


Key Concepts

It also has an “adjective problem”

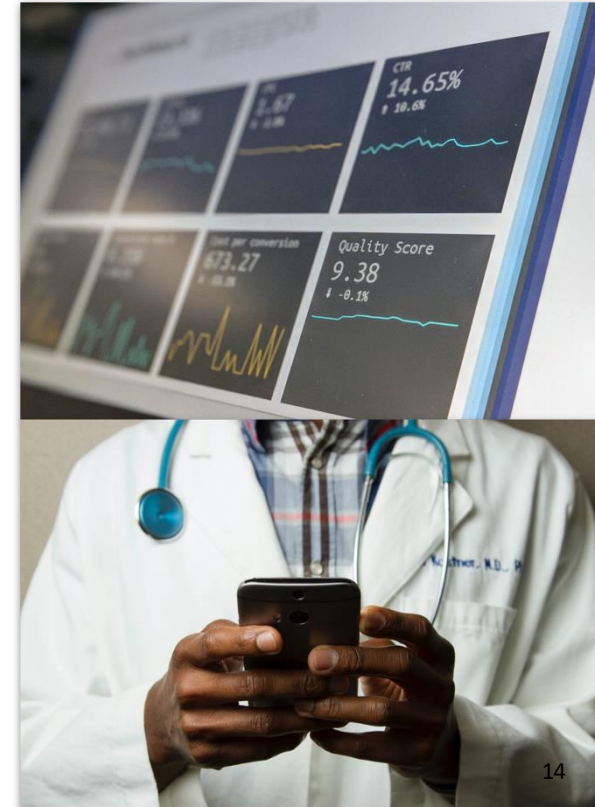
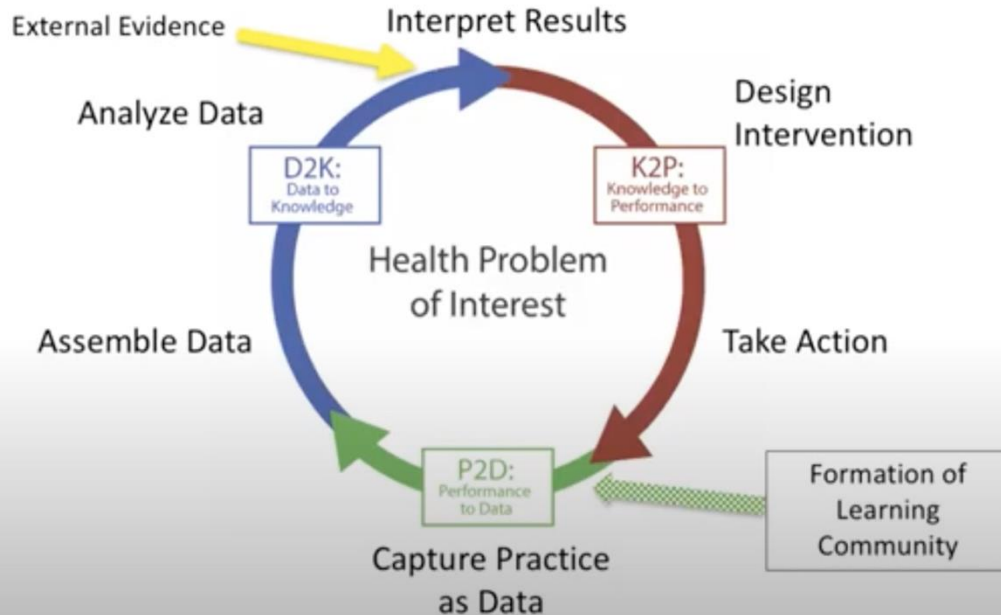


(Hersh, 2009; adapted from Shortliffe textbook)

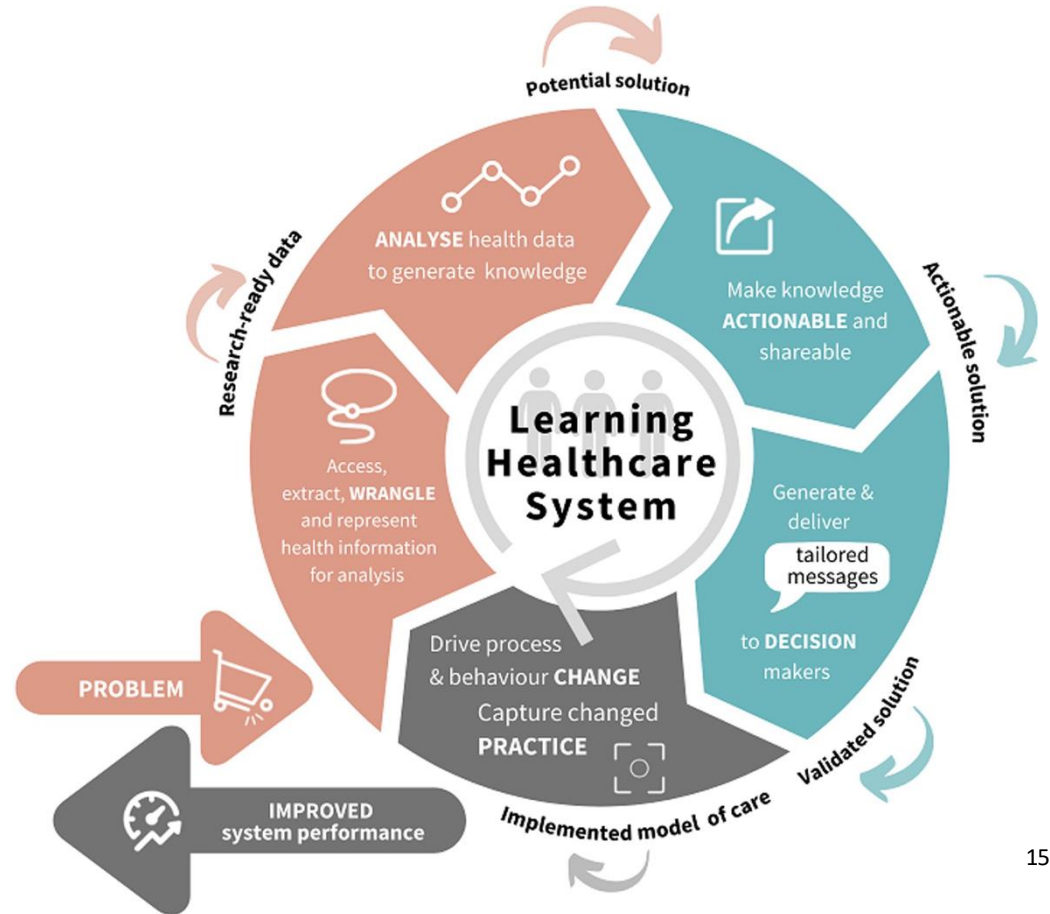
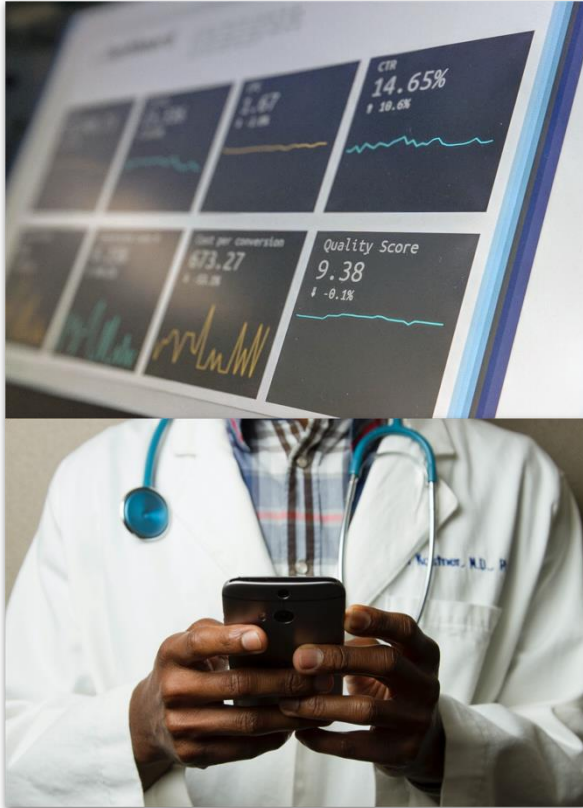


Key Concepts

The Fundamental Activity: Community-Directed Cycles of Study and Change



Open Discussion





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